

# Methods Camp

## Session 1

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# Welcome and objectives

- ▶ We are excited to welcome you to UT and to help you prep for your first year
- ▶ **Goal:** build comfort and exposure with the basic language of quantitative work you will see throughout methods courses and papers.
- ▶ We will move slowly, use sociological examples, and practice with exercises.
- ▶ Ask lots of questions (no bad questions)

# Methods Bootcamp Schedule

|              |                      |                  |          |
|--------------|----------------------|------------------|----------|
| <b>Day 1</b> | Monday, August 17    | 2:00–4:30 PM     | Math     |
| <b>Day 2</b> | Tuesday, August 18   | 9:00 AM–12:00 PM | Math     |
| <b>Day 3</b> | Wednesday, August 19 | 9:00 AM–12:00 PM | Coding   |
| <b>Day 4</b> | Thursday, August 20  | 9:00 AM–12:00 PM | Coding   |
| <b>Day 5</b> | Friday, August 21    | 9:00–10:00 AM    | Workflow |

# Why Do We Need Functions?

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  - ▶ “Poverty is negatively associated with high school graduation.”
  - ▶ “Adult mortality varies by educational attainment.”

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  - ▶ “Poverty is negatively associated with high school graduation.”
  - ▶ “Adult mortality varies by educational attainment.”
- ▶ A **function** describes **how an outcome changes as some other quantity changes**.

# Why Learn Functions?

Learning **functions** will help you understand:

- ▶ How researchers model relationships between variables
- ▶ What regression coefficients and equations mean
- ▶ How to interpret tables and figures in research articles

# What is a Function?

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- ▶ Example in words: “Take a respondent’s years of schooling, and the function tells you their predicted income.”

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- ▶  $f(x)$  is the **output** (also called the **dependent variable**) that depends on  $x$ .
- ▶ The letter  $f$  is just a label for the rule; it could be  $f$ ,  $g$ , or any name.

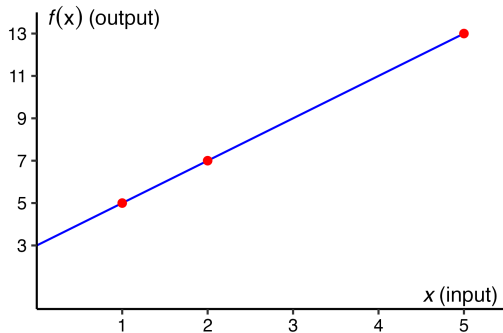
# Evaluating a Function

Let  $f(x) = 2x + 3$ .

- ▶ To evaluate the function at a specific input, substitute that number in for  $x$ .

| $x$ | Substitute | $f(x)$ |
|-----|------------|--------|
| 1   | $2(1) + 3$ | 5      |
| 2   | $2(2) + 3$ | 7      |
| 5   | $2(5) + 3$ | 13     |

# Graphing a Function



*Note: a straight line means the output increases by the same amount for every one-unit increase in  $x$  — this is what makes  $f(x) = 2x + 3$  a **linear** function.*

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- ▶  $f(x, y)$  — a rule with two inputs,  $x$  and  $y$ .
- ▶ Example in words: “a respondent’s reported life satisfaction depends on both their income *and* their health.”

# Functions With More Than One Input

$$f(x, y) = x + 2y$$

$$f(1, 4) = 1 + 2(4) = 9$$

$$f(3, 2) = 3 + 2(2) = 7$$

# Exponents and Logs

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- ▶ **Some social variables are skewed.**
  - ▶ Income and wealth may have a few very large values.
- ▶ **Logs are a shortcut for percentage change.**
  - ▶ Comes up constantly when describing change over time.

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▶  $10^2 = 10 \times 10 = 100$

▶  $b$  is the **base**;  $n$  is the **exponent**, or power.



# Exponents: Key rules

$$b^m \times b^n = b^{m+n}$$

$$\frac{b^m}{b^n} = b^{m-n}$$

$$(b^m)^n = b^{mn}$$

$$b^0 = 1, \quad b^{-n} = \frac{1}{b^n}$$

# What is a Logarithm?

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$$\log_b(x) = y \quad \Longleftrightarrow \quad b^y = x$$

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- ▶ Example:  $\log_{10}(100) = 2$  because  $10^2 = 100$ .
- ▶ Example:  $\log_2(8) = 3$  because  $2^3 = 8$ .

# Natural Log

In most statistical software, the default logarithm is the **natural logarithm**,

$$\ln(x) = \log_e(x), \quad e \approx 2.718.$$

# Log Rules

$$\ln(a \times b) = \ln(a) + \ln(b)$$

$$\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$$

$$\ln(a^n) = n \ln(a)$$

$$\ln(1) = 0, \quad \ln(e) = 1$$

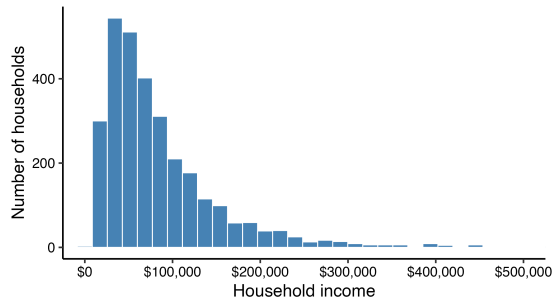
# Why Sociologists Log-Transform Variables

Household income is a classic example of a **right-skewed** variable (a few very large values).

|                      |        |        |        |        |         |
|----------------------|--------|--------|--------|--------|---------|
| Raw income (\$)      | 30,000 | 45,000 | 60,000 | 90,000 | 900,000 |
| $\ln(\text{income})$ | 10.31  | 10.71  | 11.00  | 11.41  | 13.71   |

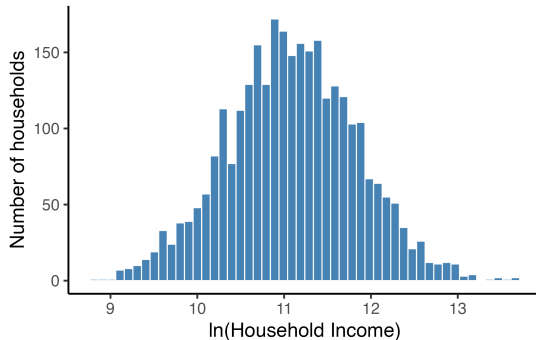
# Why Sociologists Log-Transform Variables

- ▶ On the raw scale, the last value is **wildly far** from the rest.



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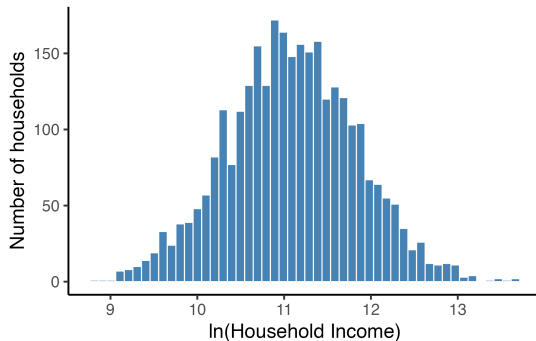
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# Why Sociologists Log-Transform Variables

- ▶ On the log scale, the gap shrinks dramatically.
- ▶ This is why many social science variables are reported or modeled “in logs”—the transformation makes extreme values far less dominant.



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- ▶  $x_i$  is the value being added for observation  $i$ .

## Summation: Example

Suppose we survey 5 respondents about hours of TV watched per day:  
 $\{2, 3, 1, 4, 5\}$ .

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To build the **sample mean**:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{15}{5} = 3$$



# Useful Summation Properties

$$\sum_{i=1}^n c = n \cdot c \quad (\text{sum of a constant})$$

$$\sum_{i=1}^n c x_i = c \sum_{i=1}^n x_i \quad (\text{constants pull out})$$

$$\sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i \quad (\text{sum of sums})$$

# The Product Operator ( $\Pi$ )

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- ▶ Example:  $\prod_{i=1}^4 i = 1 \times 2 \times 3 \times 4 = 24$

# Exercise 1: Functions and Operations

## Work in pairs.

1. Let  $f(x) = 5x - 4$ . Compute  $f(2)$  and  $f(0)$ .
2. Let  $f(x, y) = 3x + y^2$ . Compute  $f(2, 3)$ .
3. Simplify:  $\ln(50) - \ln(10)$ . (Hint: use a log rule.)
4. Solve for  $x$ :  $\ln(x) = 2$ . (Hint: rewrite in exponential form.)
5. A class survey of 6 students records number of siblings:  $\{1, 0, 2, 3, 1, 2\}$ .
  - ▶ Write this as a summation expression and compute  $\sum_{i=1}^6 x_i$ .
  - ▶ Compute the mean  $\bar{x}$ .

# Solution: Exercise 1

1.  $f(2) = 5(2) - 4 = 6; \quad f(0) = -4$
2.  $f(2, 3) = 3(2) + 3^2 = 6 + 9 = 15$
3.  $\ln(50) - \ln(10) = \ln(50/10) = \ln(5) \approx 1.609$
4.  $\ln(x) = 2 \Rightarrow x = e^2 \approx 7.389$
5.  $\sum_{i=1}^6 x_i = 1 + 0 + 2 + 3 + 1 + 2 = 9; \quad \bar{x} = 9/6 = 1.5$

# Ratios and Proportions

# What Is a Ratio?

A **ratio** compares the size of one quantity to another. It can be written  $a : b$  or  $a/b$ .



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- ▶ A ratio compares **part to part** — it does not need to reference the whole group at all.

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- ▶ Example: 18 women out of 30 total students  $\Rightarrow$  proportion female  $= 18/30 = 0.60$ .
- ▶ Unlike a ratio, a proportion always references the entire group.
- ▶ Multiply a proportion by 100 to get a **percentage**:  $0.60 \times 100 = 60\%$ .

# Rates, Relative Change, and Percentages

# Rates

A **rate** is a ratio that relates the number of events to a population, time period, or another measure of exposure, allowing meaningful comparisons across groups.



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A **rate** is a ratio that relates the number of events to a population, time period, or another measure of exposure, allowing meaningful comparisons across groups.

► **Example: Crime rate per 100,000 people**

$$\text{Crime rate} = \frac{\# \text{ crimes during a given period}}{\text{population at risk}} \times 100,000$$

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**rates let us compare *risk*, not just volume.**

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A **percentage** is simply a proportion expressed on a 0–100 scale instead of a 0–1 scale.

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This tells the raw size of the change, but not whether that change is large or small **relative to where you started**.

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This tells the size of the change **relative to the starting point** — usually the more meaningful number for comparing across groups.

# Percentage Points vs. Percent Change

**Example:** The unemployment rate rises from **5%** to **7%**.

► **Percentage point change:**

$$7 - 5 = 2 \text{ percentage points}$$

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► **Percent change:**

$$\frac{7 - 5}{5} \times 100 = 40\%$$

# Percentage Points vs. Percent

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# Percentage Points vs. Percent

- ▶ A **percentage point** is the arithmetic difference between two percentages.
- ▶ A **percent change** is the relative difference, expressed as a percentage of the starting value.
- ▶ These are **not** the same number, and mixing them up **changes the meaning** of a claim.

# Recap: Rates, Change, and Percentages

- ▶ **Rate**: a ratio standardized by a base (often population and/or time) so groups can be compared fairly.
- ▶ **Percentage**: a proportion  $\times 100$ .
- ▶ **Absolute change**:  $\text{New} - \text{Old}$ .
- ▶ **Percent (relative) change**:  $\frac{\text{New} - \text{Old}}{\text{Old}} \times 100$ .
- ▶ **Percentage point**: the raw arithmetic gap between two percentages — **not** the same as percent change.

## Exercise 2: Ratios, Proportions, Rates & Percentages

A town has 5,400 women and 4,600 men.

1. Compute the sex ratio (males per 100 females).
2. Compute the proportion of the town that is female.
3. Last year the town recorded 27 burglaries. Compute the burglary rate per 100,000 residents. (Total population = 10,000.)
4. The town's unemployment rate fell from 8% to 6%.
  - ▶ What is the change in **percentage points**?
  - ▶ What is the **percent (relative) change**?
5. A local news headline says: "Unemployment drops 2%!" Is this headline accurate? Why or why not?

## Solution: Exercise 2

1. Sex ratio =  $\frac{4,600}{5,400} \times 100 \approx 85.2$  males per 100 females.
2. Proportion female =  $\frac{5,400}{10,000} = 0.54$  (54%).
3. Burglary rate =  $\frac{27}{10,000} \times 100,000 = 270$  per 100,000 residents.
4. Percentage-point change:  $6 - 8 = -2$  p.p.    Percent change:  
 $\frac{6 - 8}{8} \times 100 = -25\%$ .
5. The headline is **imprecise**: the rate fell by 2 *percentage points*, not 2 percent. The true relative decline is 25%.

## Exercise 3

A researcher studies how income relates to years of education ( $X$ ), using a model of **log income**:

$$\ln(Y) = 9.00 + 0.08 X$$

where  $Y$  is annual income in dollars.

1. For a respondent with  $X = 12$  years of education, compute  $\ln(Y)$ .
2. Convert  $\ln(Y)$  into  $Y$  (dollars) by exponentiating. (Hint: use the inverse of  $\ln$ .)
3. Now consider a respondent with  $X = 13$ . Compute their  $\ln(Y)$  and their  $Y$ .
4. Exponentiate the coefficient on  $X$ :  $e^{0.08} \approx 1.083$ . Interpret this number in one sentence.

## Solution: Exercise 3

1.  $\ln(Y) = 9.00 + 0.08(12) = 9.00 + 0.96 = 9.96$
2.  $Y = e^{9.96} \approx \$21,159$
3.  $\ln(Y) = 9.00 + 0.08(13) = 10.04$ ;  $Y = e^{10.04} \approx \$22,926$
4.  $e^{0.08} \approx 1.083$ : each additional year of education is associated with income being multiplied by about **1.08** — roughly an **8% increase** in income, *holding other factors constant*.